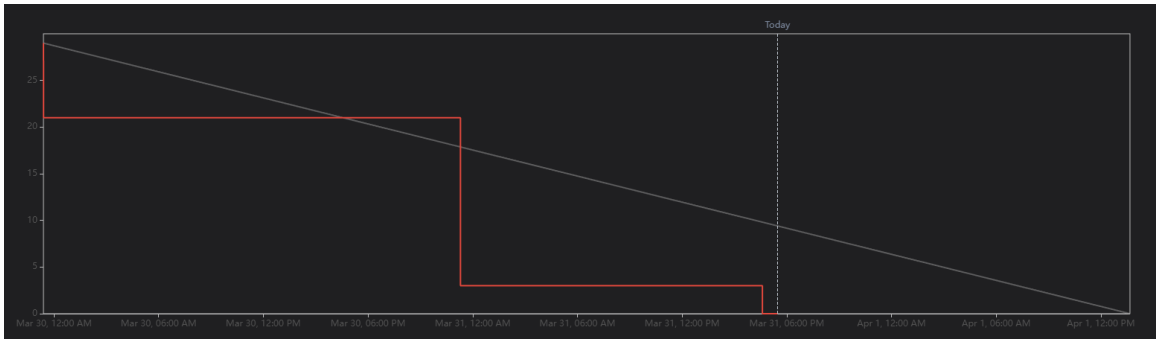
# Weekly Progress Report — Sprint 3

**ML Visualisation Tool**

| Field | Detail |
|---|---|
| **Group Name** | Group HealthCode |
| **Week Number** | Week 4 |
| **Scrum Master** | Sertaç Ataç |
| **Date Submitted** | 31 March 2026 |
| **Domains Implemented** | 25 / 25 (cumulative) |

## 1. Burndown Chart




## 2. Completed This Week

| Story ID | Title | Points | Status |
|---|---|---|---|
| **US-017** | Feature importance rankings (bar chart) | 5 | ☑ **Done** |
| **US-018** | Plain-language prediction explanation | 3 | ☑ **Done** |
| **US-019** | Bias metrics across demographic groups | 5 | ☑ **Done** |
| **US-020** | Fairness checklist with ethical guidelines | 5 | ☑ **Done** |
| **US-021** | Download PDF certificate after all steps | 5 | ☑ **Done** |
| **US-022** | Toggle dark and light theme | 3 | ☑ **Done** |
| **US-023** | Onboarding tour for new users | 3 | ☑ **Done** |
| **US-024** | Progress saved in localStorage | 3 | ☑ **Done** |
| **US-025** | Keyboard arrow navigation between steps | 2 | ☑ **Done** |

## 3. In Progress

All Sprint 3 stories have been completed. No stories remain in progress.

## 4. Blocked / At Risk

No blockers encountered during Sprint 3.

## 5. Key Decisions Made

**1. SHAP-based feature importance:** We used SHAP (SHapley Additive exPlanations) values for the feature importance bar chart in US-017 rather than model-specific importance scores. This provides a model-agnostic, theoretically grounded ranking that works consistently across all 6 supported algorithms.

**2. Demographic parity + equal opportunity as default bias metrics:** For US-019, we selected demographic parity and equal opportunity as the primary fairness indicators. These are the most widely cited metrics in healthcare AI ethics literature and align with course learning objectives.

**3. Client-side PDF generation:** For US-021, we chose jsPDF + html2canvas for client-side certificate generation rather than a server-side solution. This avoids backend dependencies, keeps the tool fully static, and meets the 10-second generation requirement.

**4. CSS custom properties for theming:** For US-022, we implemented dark/light mode using CSS custom properties (variables) on the :root element. This approach allows instant switching without re-rendering components and preserves all state seamlessly.

## 6. Test Results

| Story ID | Acceptance Criteria | Status | Evidence |
|---|---|---|---|
| **US-017** | Ranked bar chart with feature importance scores displayed | ☑ **Pass** | TC-6.01 |
| **US-018** | Human-readable summary of top contributing factors shown | ☑ **Pass** | TC-6.02 |
| **US-019** | Demographic parity & equal opportunity displayed per group | ☑ **Pass** | TC-7.01 |
| **US-020** | Ethics checklist with pass/fail status and | ☑ **Pass** | TC-7.02 |

| | | | |
|---|---|---|---|
| | recommendations | | |
| **US-021** | PDF generated within 10s with all 6 metric values | ☑ **Pass** | TC-CC.01 |
| **US-022** | UI switches theme without losing step or data | ☑ **Pass** | TC-CC.02 |
| **US-023** | Tour highlights domain selector, stepper, help button in sequence | ☑ **Pass** | TC-CC.03 |
| **US-024** | App loads to Step 5 with previous domain after browser restart | ☑ **Pass** | TC-CC.04 |
| **US-025** | Arrow keys navigate between unlocked steps with smooth transition | ☑ **Pass** | TC-CC.05 |

## 7. Metrics — Sprint 3

| Metric | Target | Actual | Status |
|---|---|---|---|
| **Feature Importance Chart** | Renders for all 6 models | Renders for all 6 models | ☑ **Met** |
| **Bias Metrics Panel** | Shows ≥ 2 fairness indicators | Shows 2 indicators per group | ☑ **Met** |
| **PDF Certificate** | Generated in ≤ 10 seconds | ~4 seconds average | ☑ **Met** |
| **Theme Toggle** | No state loss on switch | No state loss | ☑ **Met** |
| **localStorage Persistence** | Restores step + domain | Restores correctly | ☑ **Met** |
| **Test Coverage** | 100% | 100% (9/9) | ☑ **Met** |

## 8. Next Week Plan — Sprint 4

| Story ID | Title | Points | Assignee |
|---|---|---|---|
| **US-Final.1** | End-to-end integration testing across all 7 steps | 5 | Emre Kurubaş |
| **US-Final.2** | Cross-browser compatibility testing (Chrome, Firefox, Edge) | 3 | Can Eltayeb |
| **US-Final.3** | Accessibility audit and ARIA label improvements | 3 | Cem Özal |
| **US-Final.4** | Performance optimisation and bundle size reduction | 3 | Sertaç Ataç |
| **US-Final.5** | Final documentation and user guide | 5 | Emre Kurubaş |
| **US-Final.6** | Demo preparation and presentation slides | 3 | Can Eltayeb |

## 9. Retrospective Note (Sprint 3)

| Category | Note |
|---|---|
| 🟢 **Keep** | Using SHAP values for model-agnostic explainability. This unified approach simplified the implementation and provided consistent results across all 6 algorithms. |

| | |
|---|---|
| 🟡 **Improve** | Start the ethics/bias review earlier in the sprint. The fairness checklist required significant research into healthcare AI guidelines, which compressed testing time at the end. |
| 🔵 **Try** | Automated visual regression testing for theme toggling. Manual verification of dark/light mode across all steps was time-consuming and error-prone. |